Clinical Diagnosis of Leprosy

Skin lesions in leprosy are typically hypopigmented or erythematous macules, papules, nodules, or plaques, which may be skin-colored or slightly red. Clinically, leprosy is diagnosed when a patient presents with at least two of the three cardinal signs. However, the WHO considers a single cardinal sign sufficient for diagnosis in endemic areas, often overlooking nerve enlargement.

The Three Cardinal Signs of Leprosy (Known for Over 100 Years):

1. **Loss of sensation in a skin lesion**
2. **Enlarged peripheral nerve**
3. **Positive skin smears**

To date, there are no definitive laboratory tests to diagnose leprosy. Available tests support classification, monitoring, and detection of relapse but are not diagnostic on their own.

1. Loss of Sensation

Loss of sensation is assessed using a wisp of cotton wool. The lesion is touched—**not brushed**—while the patient's eyes are closed. The patient indicates where they feel the touch. To confirm accuracy, sensation is also tested in the skin surrounding the lesion (Fig. 2).

Alternative sensory tests include: - Testing for warm and cold perception - Assessing diminished sweating (e.g., iodide starch test) - Histamine test on the affected area compared with normal skin (evaluates axonal reflex)

These tests assess damage to dermal nerves. Notably, leprosy primarily affects the nervous system, making it fundamentally a **neural disease of the skin**.

2. Enlarged Peripheral Nerves

Enlarged nerves may involve: - Cutaneous nerves - Subcutaneous nerves near skin lesions - Major nerve trunks

Key nerves to palpate include: - Posterior auricular nerves - Ulnar nerves - Radicutaneous nerves - Median nerves - Lateral popliteal nerves - Posterior tibial nerves

Assessment should include evaluation of: - Nerve thickness - Consistency - Tenderness (Fig. 3a,b)

Ultrasound is a valuable tool for assessing nerve enlargement.

It is important to note that enlarged fasciculi within the nerve may present a characteristic diagnostic pattern. Clinicians should also check for sensory loss or muscle weakness in the distribution of the affected nerve, although these signs may not always be apparent. Nerve conduction studies can provide additional diagnostic support.

3. Skin Smears for Acid-Fast Bacilli

Smears are collected from: - Earlobes and other cooler body areas - The **edge of the lesion** in paucibacillary (PB) patients - The **center of the lesion** in multibacillary (MB) patients

Technique: - Squeeze the skin to reduce bleeding and numb the area - Make a small incision into the dermis - Collect tissue fluid only—**blood dilutes the sample** and reduces bacillary concentration

Bacilli are quantified using: - **Bacillary Index (BI):** Logarithmic scale for bacilli count - **Morphological Index (MI):** Percentage of intact (viable) bacilli

Staining Considerations:

- Use **1% hydrochloric acid in isopropyl alcohol** for decolorization (as in Fite stain)

- Avoid **3% hydrochloric acid** (used in Ziehl-Neelsen stain for TB), as *M. leprae* is less acid-fast than *M. tuberculosis*
- Using 3% acid may result in false-negative smears

Molecular Detection:

- PCR and NASBA can detect *M. leprae* DNA/RNA
- However, these methods are often negative in PB patients
- Smears, immunological assays, and molecular techniques are most useful in diagnosing MB leprosy, monitoring treatment, and detecting relapse

Additional Laboratory Investigations

Other laboratory tests may support the diagnosis but are **not universally diagnostic** across the spectrum of leprosy. No single test confirms leprosy in all cases. Clinical evaluation remains paramount.